

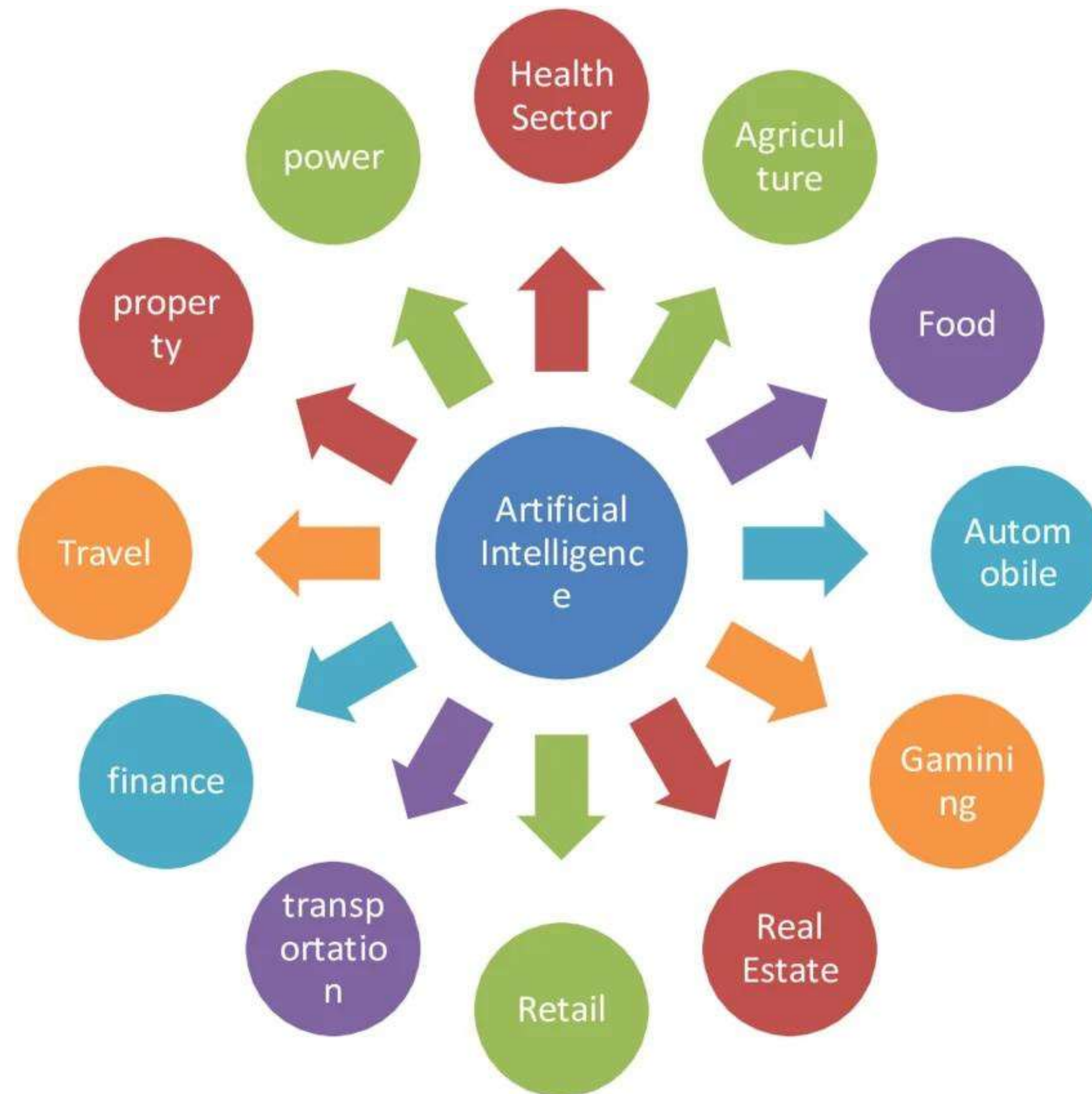
AI Unlocked: The Fundamentals

Presented
by :

MOHAMED CHEBBI

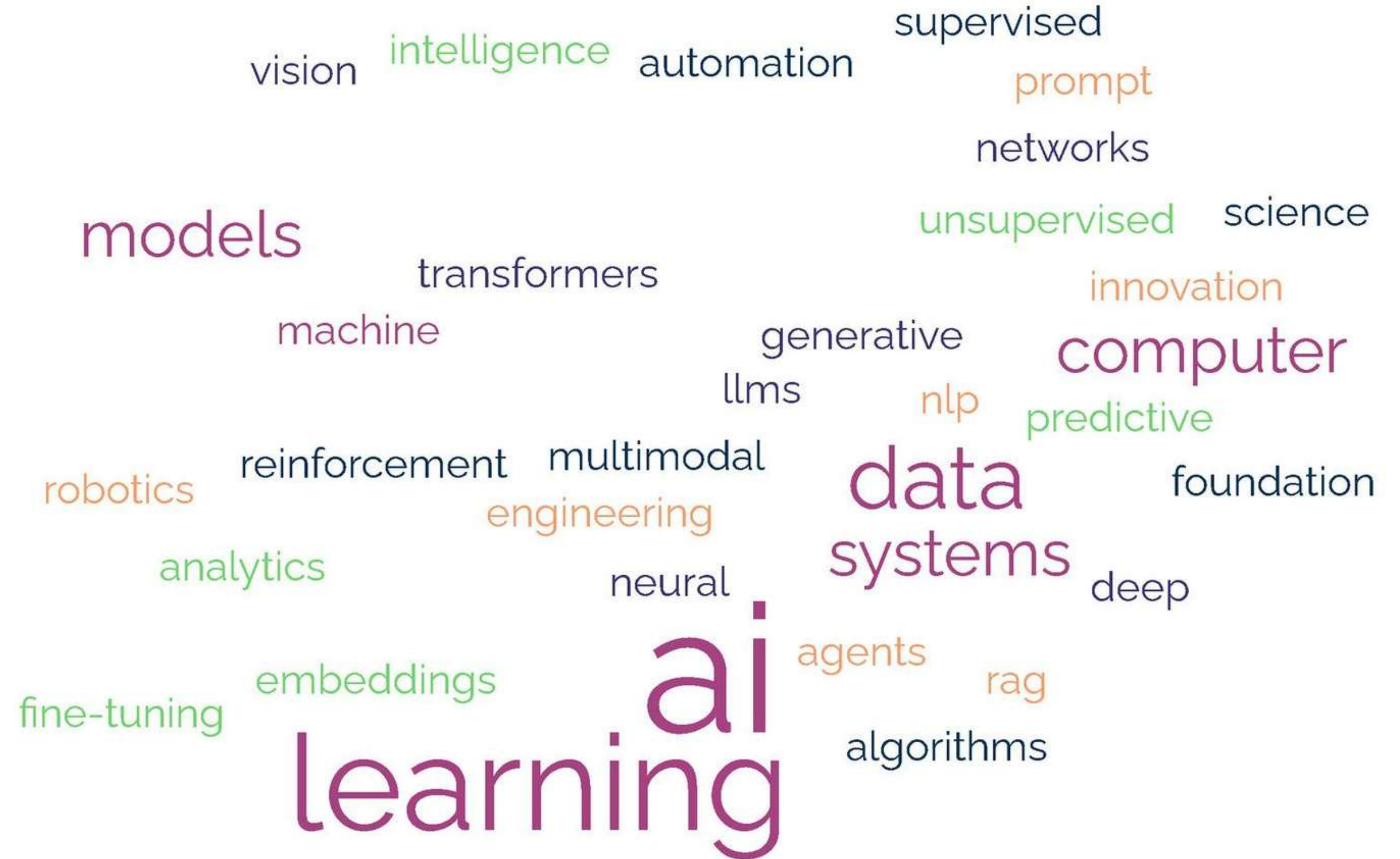


AI Fields of Application:





Artificial Intelligence?



History of AI: Turing's Foundations

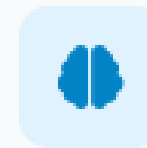
Alan Turing laid the foundations of AI as early as **1948–1950** through two core pillars:



1948 Foundation

Ancestors of Neural Networks

Theorized "***unorganized machines***" learning through trial and error (reinforcement learning principle).



1950 Foundation

The Child Brain Concept

Proposed simple systems that ***learn from data and experience*** rather than hand-coded human knowledge.

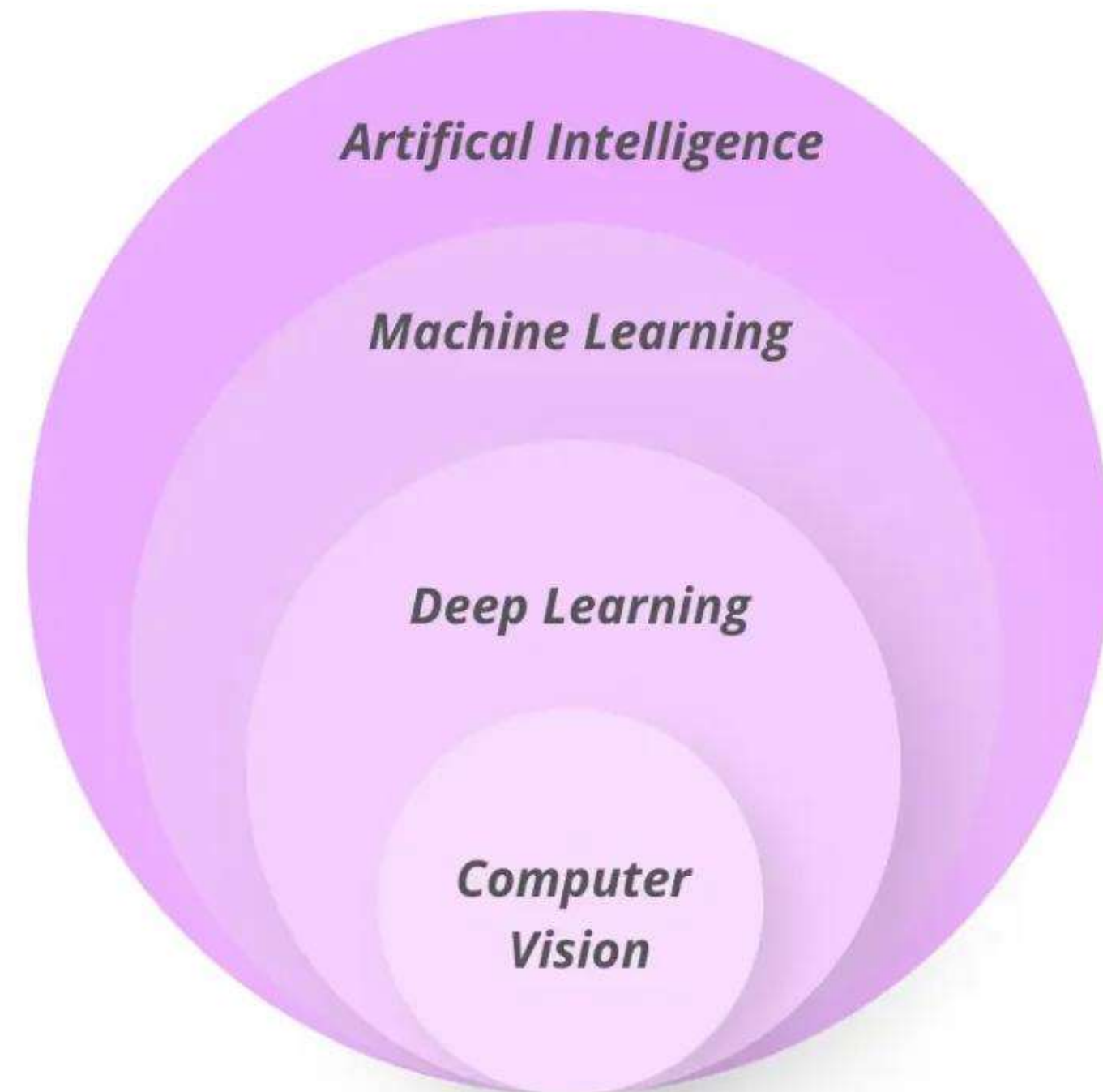
Robot seem as single AI system but several AI systems run in parallel !



- ***Environment Mapping System → Computer Vision***
- ***Object Identification System → Deep Learning***
- ***Grip & Movement Control System → Machine Learning***

Hierarchy of Artificial Intelligence

AI and Its Subfields



THE PRE-STATISTICAL ERA

Symbolic NLP

How we processed language before machine learning and deep neural networks.

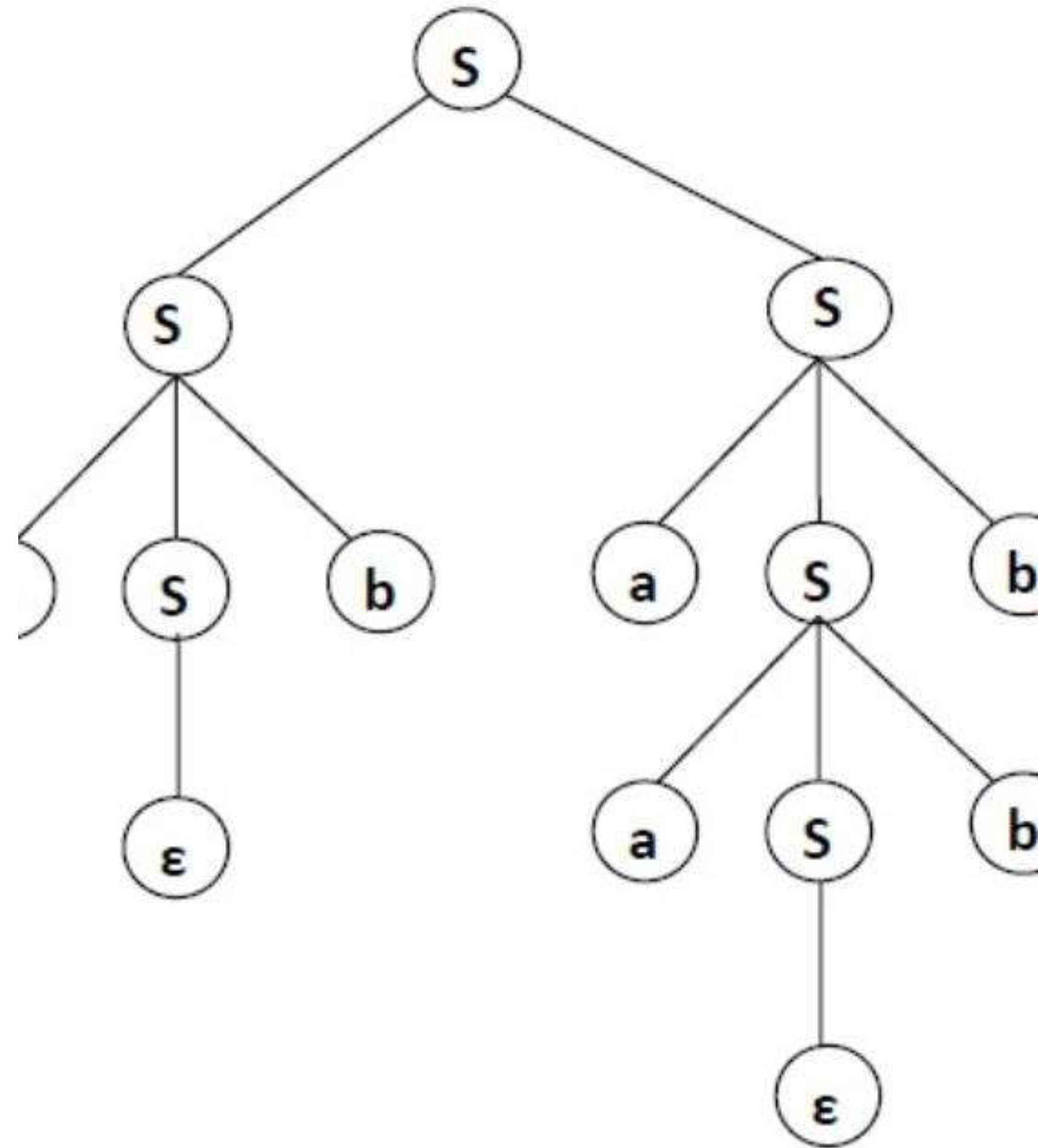


Parsing Syntax

Language mapped to logic.

Noun and verb phrases.

Rigid mathematical equations.



The Rule-Based Pipeline



Pipeline in Action

1. Tokenization

Input String:

"The robot
speaks."

Output Tokens:

["The"]

["robot"]

["speaks"]

["."]

2. Parsing

Syntactic Structure:

S |—— NP (Noun Phrase) | |—— Det: "The" | |—— N: "robot" |—— VP (Verb Phrase) |—— V:
"speaks" |—— Punc: "."

The Rule-Based Tradeoff

Strengths

Perfectly explainable logic.

Zero training data required.

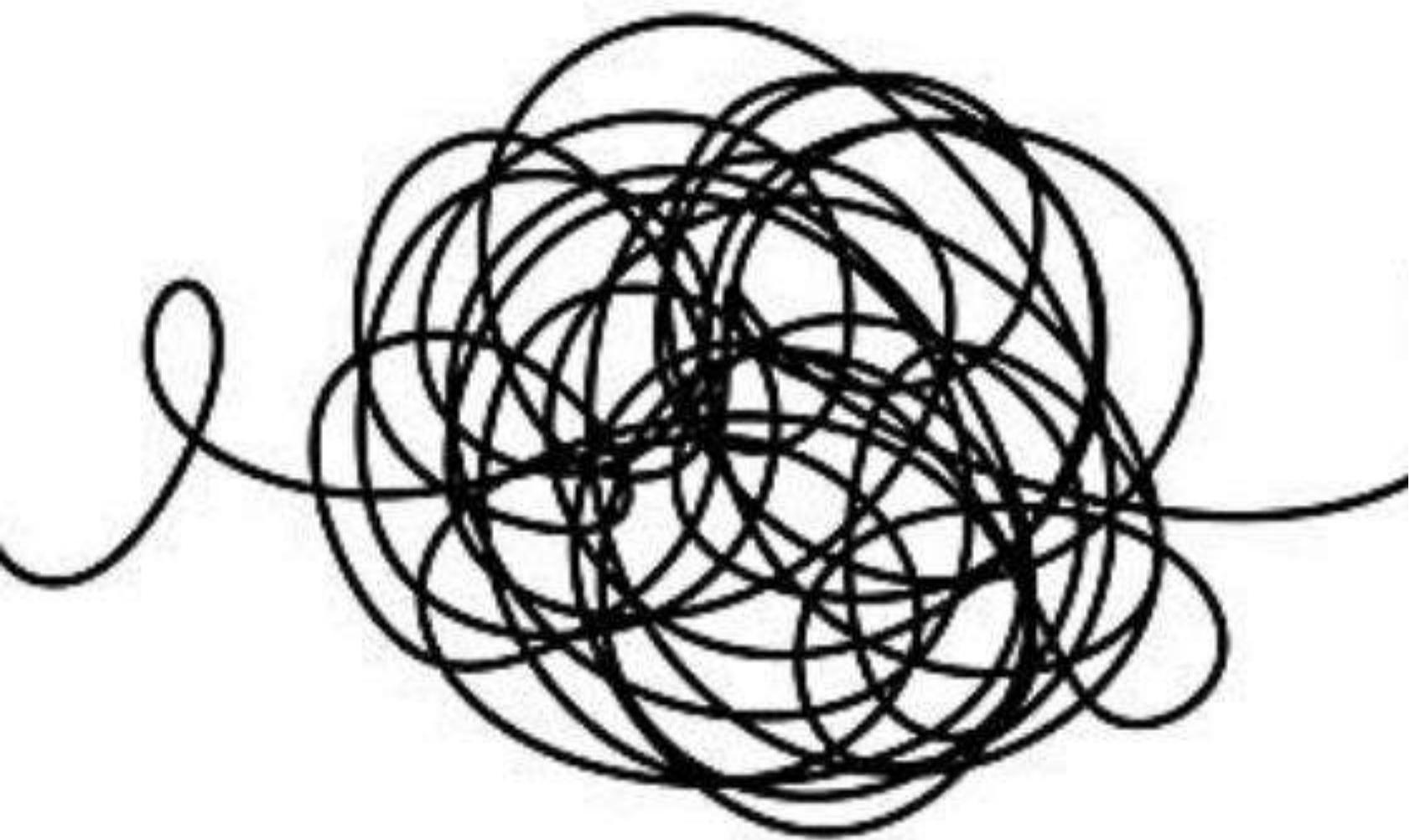
High precision in niches.

Weaknesses

Brittle edge cases.

Combinatorial rule explosion.

Expensive expert labor.



The Ambiguity Wall

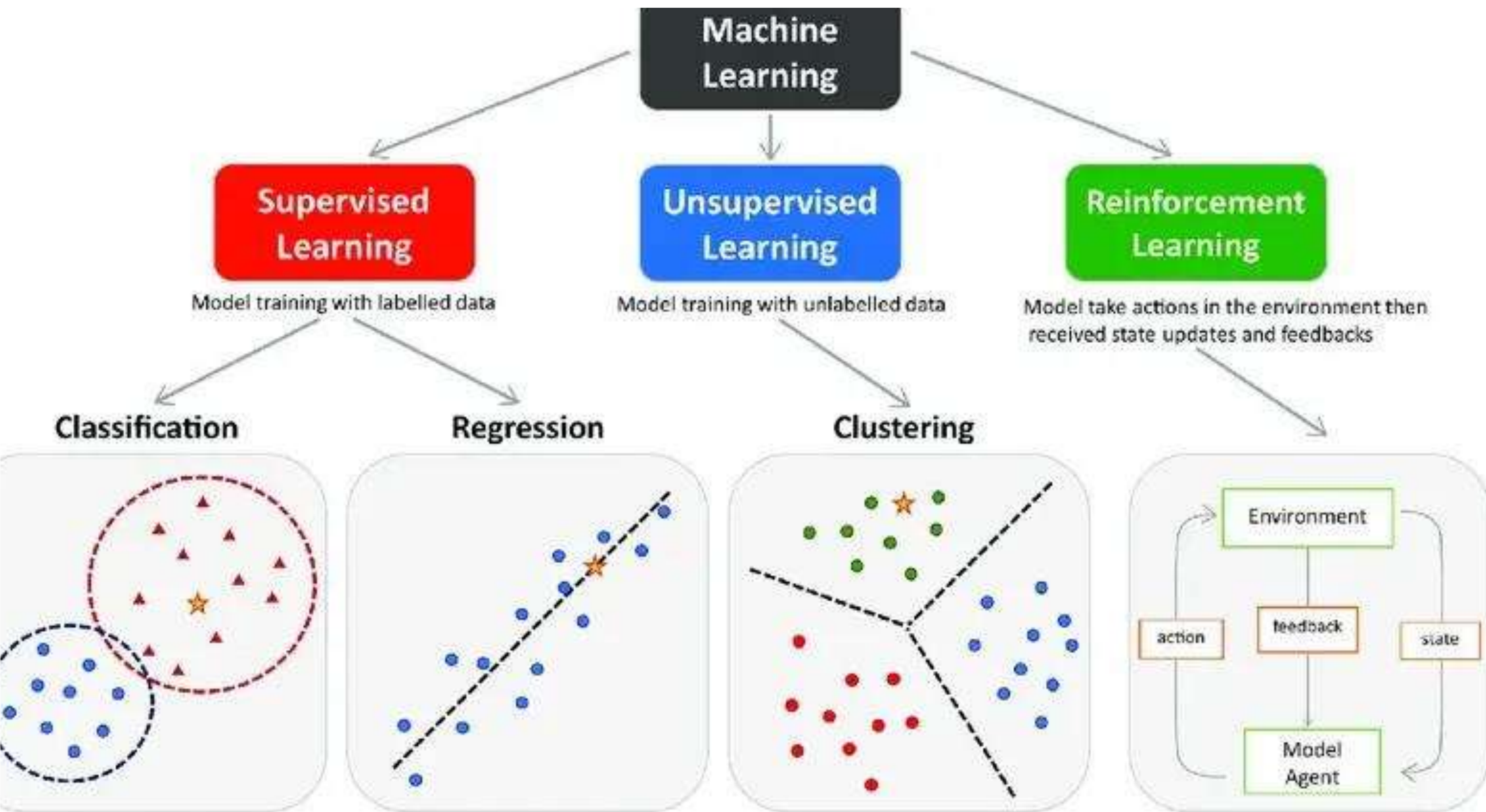
Language is inherently messy.

Sarcasm breaks logic rules.

Context changes fixed meanings.

An Overview of Machine Learning (Core Engine)

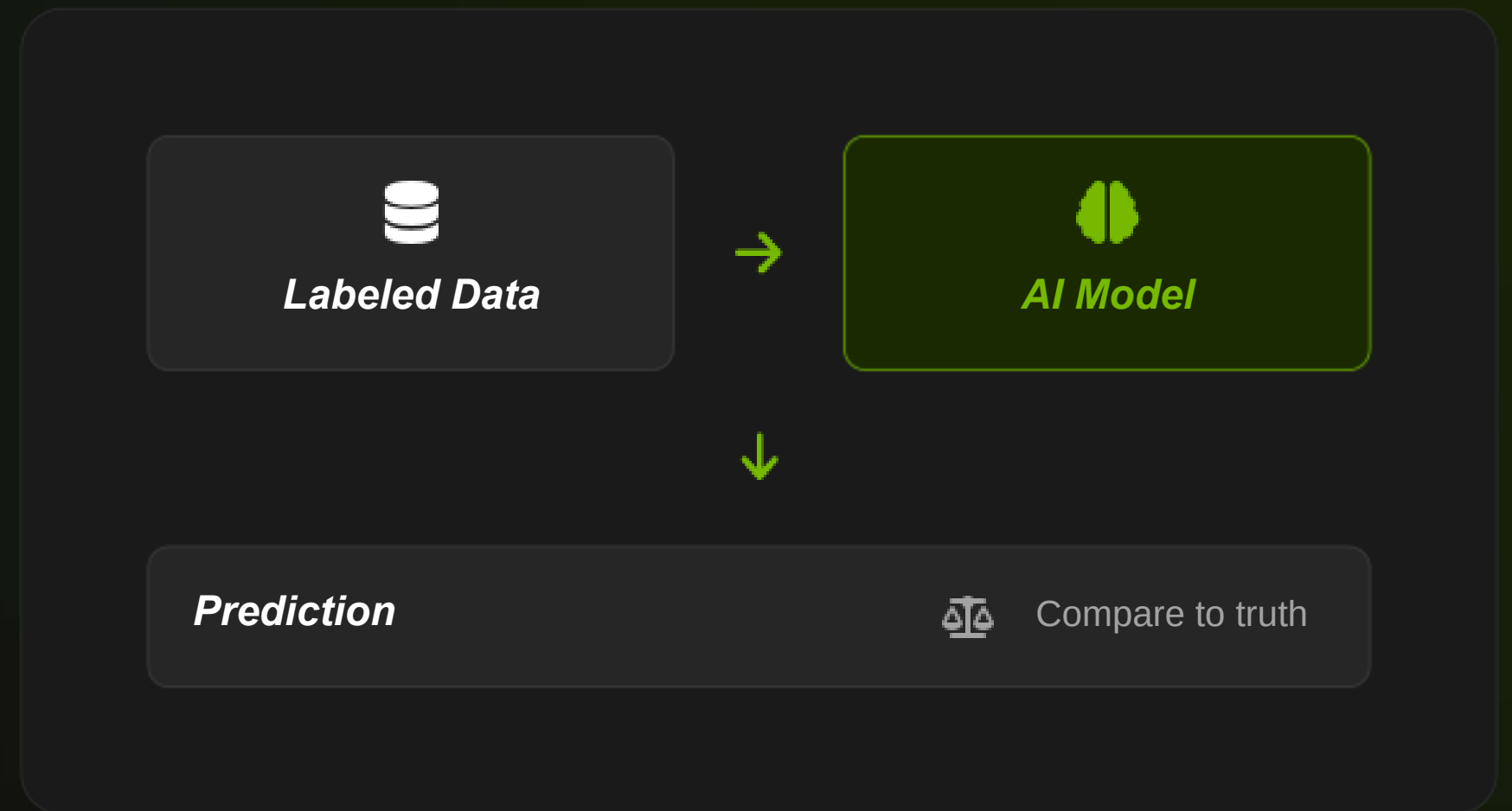
You can think of machine learning as ***training a model to make decisions from data***. Instead of relying only on ***fixed, handwritten rules***, the model ***learns patterns from examples*** and improves as it is ***trained and evaluated***. Once trained, it can ***make predictions, classify inputs, find trends, or flag unusual cases*** based on what it learned.



What is Supervised

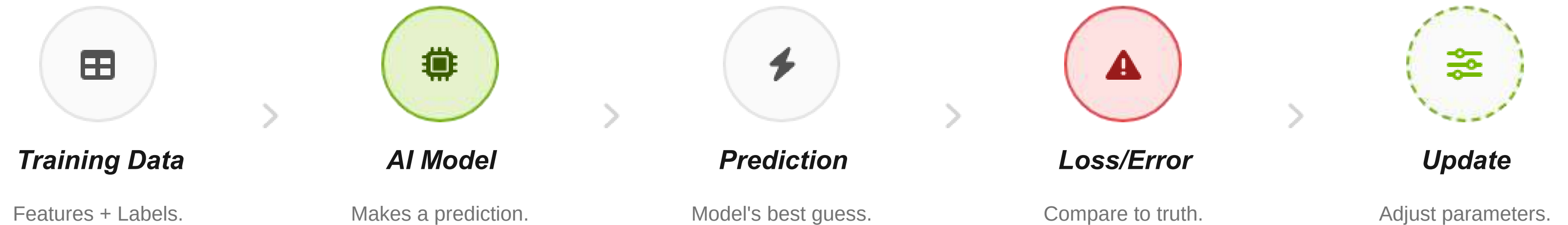
Learning with a teacher. Training a model
using examples paired with their correct
answers.

Learning?



THE MECHANICS

How Supervised Learning Works



Process repeats iteratively until the model learns useful patterns.

Two Major Branches

CLASSIFICATION

Predicts a category.

Input → ***[Class A | Class B]***

- ✓ ***Spam vs Not Spam***
- ✓ ***Fraud vs Legitimate***
- ✓ ***Disease vs No Disease***

REGRESSION

Predicts a numerical value.

Input → ***[72.4]***

- ✓ ***House price prediction***
- ✓ ***Sales forecasting***
- ✓ ***Temperature prediction***

Example: Spam Detection

Learning patterns from labeled examples.

"Congratulations! You won a prize!"

SPAM

"Meeting scheduled for 10 AM tomorrow."

NOT SPAM

"Claim your exclusive reward now!"

SPAM

NEW UNSEEN EMAIL

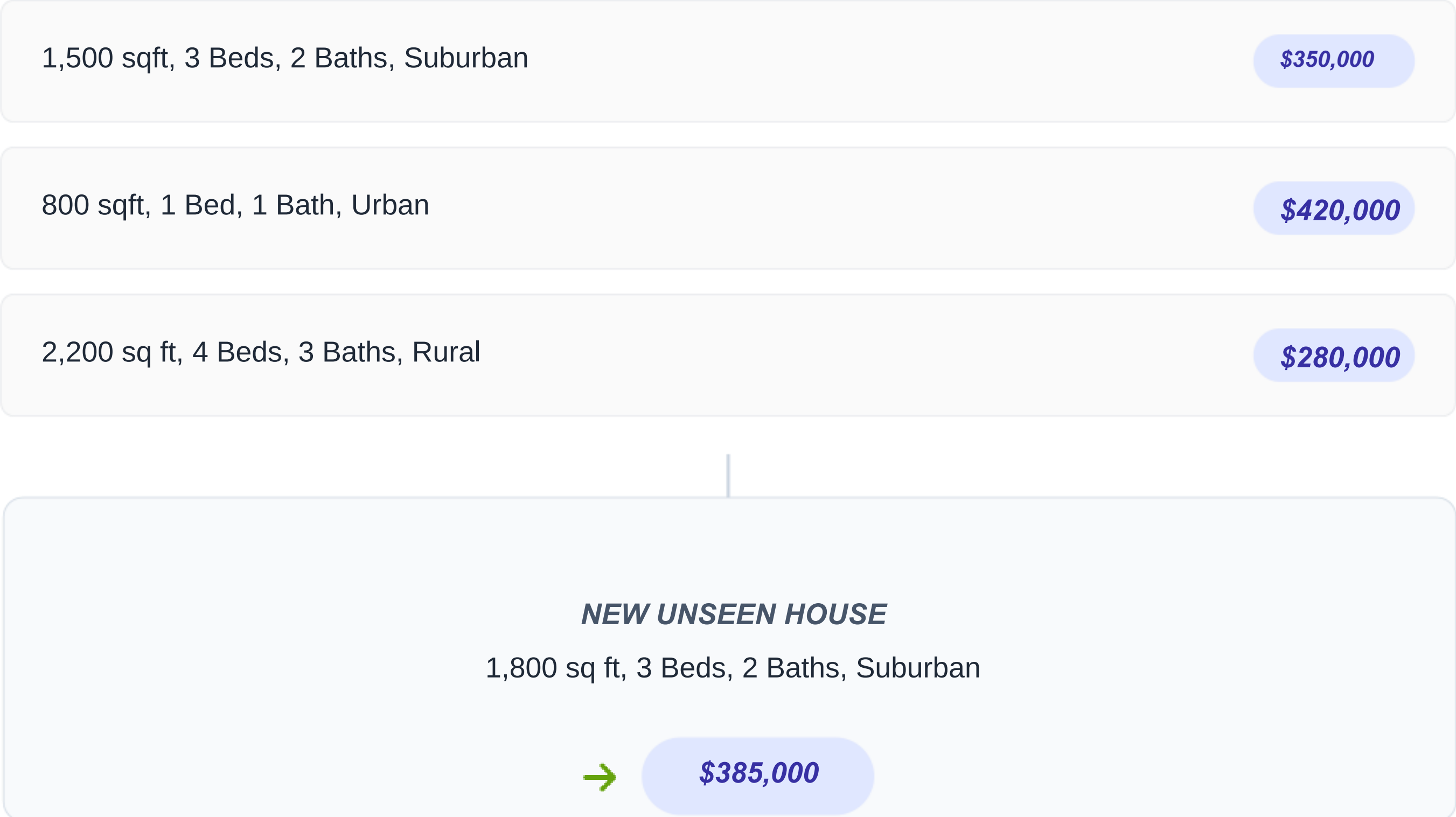
"URGENT: Click here to claim your gift!"



SPAM

Example: House Price Prediction

Predicting continuous numerical values from input features.



How Does the Model Learn?

$$W_{\text{new}} = W_{\text{old}} - \eta \cdot \nabla L$$

Weights & Bias

Feature influence and adjustable baseline.

Loss Function (L)

Measures prediction error.

Gradient Descent (∇L)

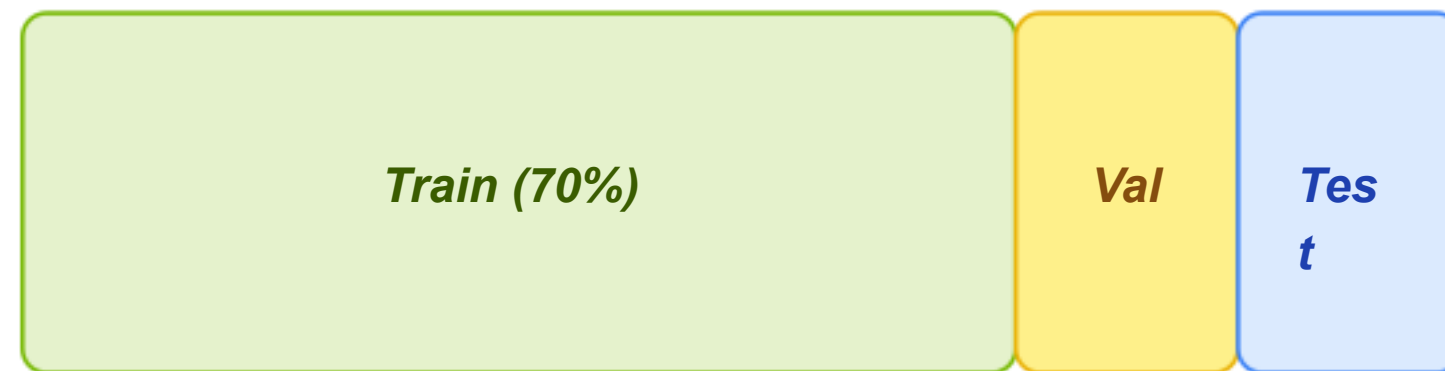
Direction to reduce the error.

Backpropagation

Calculates parameter contribution to error.

Training vs Testing

Dataset Split



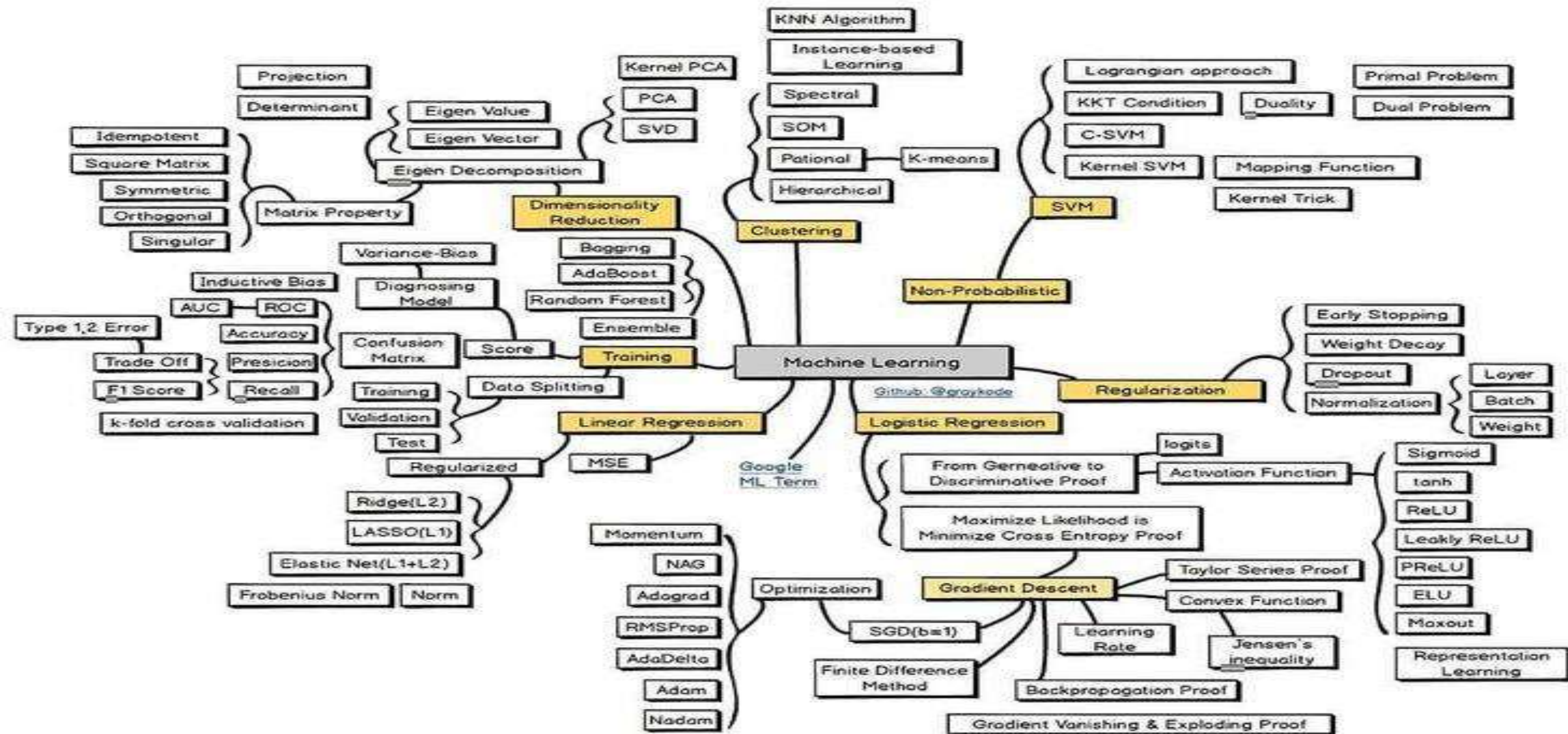
- **Training set:** Learns the patterns.
- **Validation set:** Tunes model settings.
- **Test set:** Evaluates on unseen data.



Memorization $\neq$ Generalization

A good model doesn't simply memorize the training examples—it learns patterns that work on new data.

Machine Learning: Mind Map



The Supervised Toolkit

Traditional ML and Neural Networks.

Linear Regression

REGRESSION

Logistic Regression

CLASSIFICATION

Decision Trees

BOTH

***Support Vector Machines
(SVM)***

BOTH

k-Nearest Neighbors (k-NN)

BOTH

Neural Networks

BOTH

Linear Regression

The foundational algorithm for predicting continuous values.

Core Mechanism

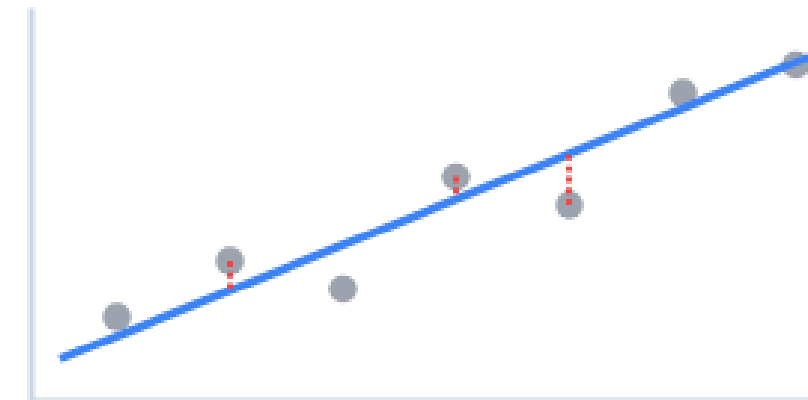
Fits a straight line to observed data points.

$$y = wx + b$$

REGRESSION

Line of Best Fit

Minimizes the distance to actual targets.



SUPERVISED

Method of Least Squares

How linear regression finds the "best fit" line.

1. Find the Errors

Measure the vertical distance between real data points and the prediction line.

REAL VALUE (Y) VS PREDICTED ($\hat{Y}$)

RESIDUALS

2. Square the Errors

Make all errors positive and strongly penalize large mistakes.

$$\Sigma (y - \hat{y})^2$$

PENALTY FUNCTION

3. Minimize the Sum

Adjust the line's slope and intercept until the total squared error is as small as possible.

OPTIMIZATION

Logistic Regression

Despite the name, it's a foundational algorithm for classification.

Binary Outcomes

Classifies data into distinct categories rather than continuous values.

Email Processing

Spam / Not Spam

Medical Tests

Positive / Negative

Transaction Risk

Fraud / Legit

CLASSIFICATION

Probability Output

Returns a likelihood score strictly between 0 and 1.

87%

PROBABILITY OF CLASS A

PROBABILITY

The Sigmoid Function

Squashing infinitesimal predictions into a bounded probability curve.

The S-Curve Math

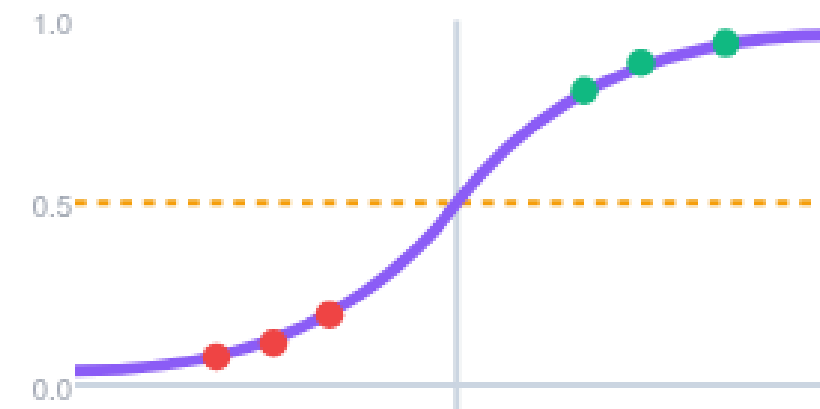
Maps any real-valued number into the (0, 1) range.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

ACTIVATION

Decision Threshold

A cutoff (usually 0.5) determines the final predicted class.



THRESHOLD

k-Nearest Neighbors (k-NN)

Predicting outcomes based on spatial proximity in feature space.

Distance as Similarity

Assumes similar data points exist in close proximity to each other.

Euclidean

Straight-line distance

Manhattan

Grid-like distance

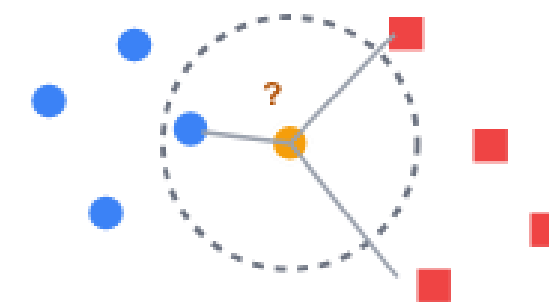
Cosine

Angle-based similarity

INSTANCE-BASED

The Majority Vote

The hyperparameter 'k' determines how many neighbors influence the prediction.



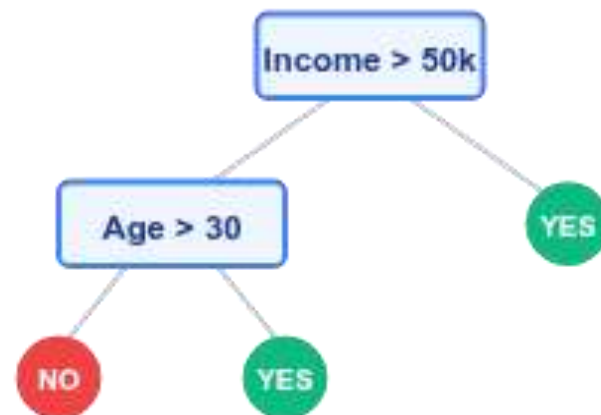
NON-PARAMETRIC

Decision Trees

Learning through a sequential hierarchy of if-then-else rules.

Recursive Splitting

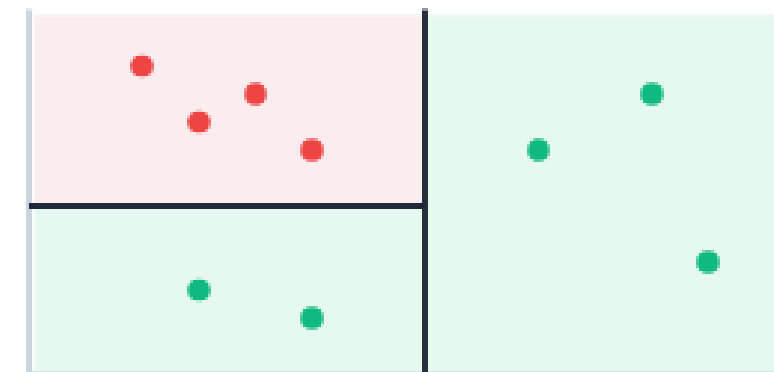
Divides data continuously to maximize purity at each new node.



RULE-BASED

Non-Linear Boundaries

Creates rigid, box-like decision regions in the feature space.



INTERPRETABLE

What is Unsupervised Learning?

Unsupervised learning discovers patterns and structure in data without predefined labels.

SUPERVISED LEARNING

Learning with a Teacher

Input + Correct Answer → Model → Prediction

UNSUPERVISED LEARNING

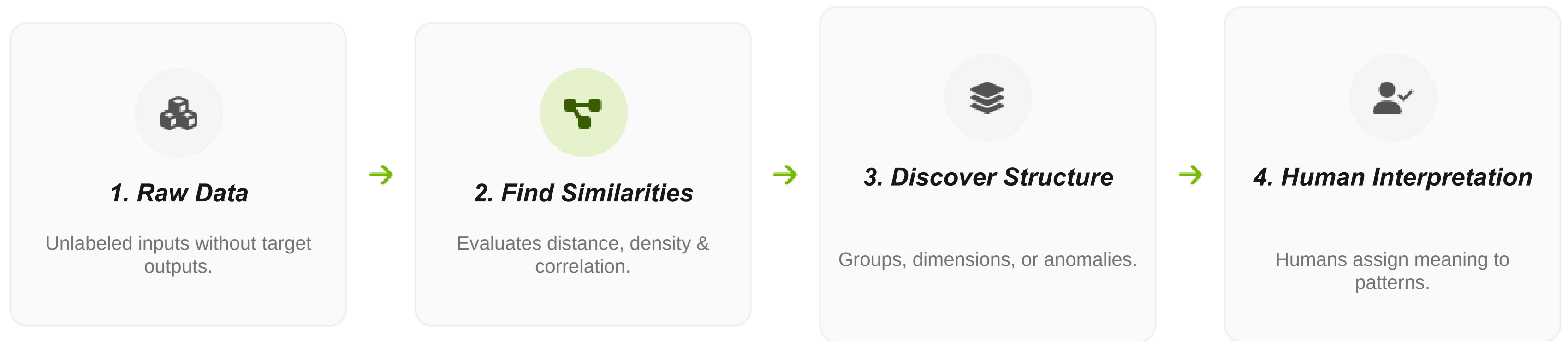
Self-Guided Discovery

Raw Unlabeled Data → Model → Discovered Structure



Give the model data without labels → let it discover hidden structure on its own.

How Unsupervised Learning Works



The algorithm discovers mathematical structure —humans interpret what that structure represents.

The Four Main Tasks

Core capabilities of unsupervised models.



1. Clustering

"Which data points belong together?"

Groups customers by spending behavior.



2. Dimensionality Reduction

"Can we simplify high-dimensional data?"

Compresses 100 features into 2D plots.



3. Anomaly Detection

"Which observations look unusual?"

Flags suspicious credit card charges.



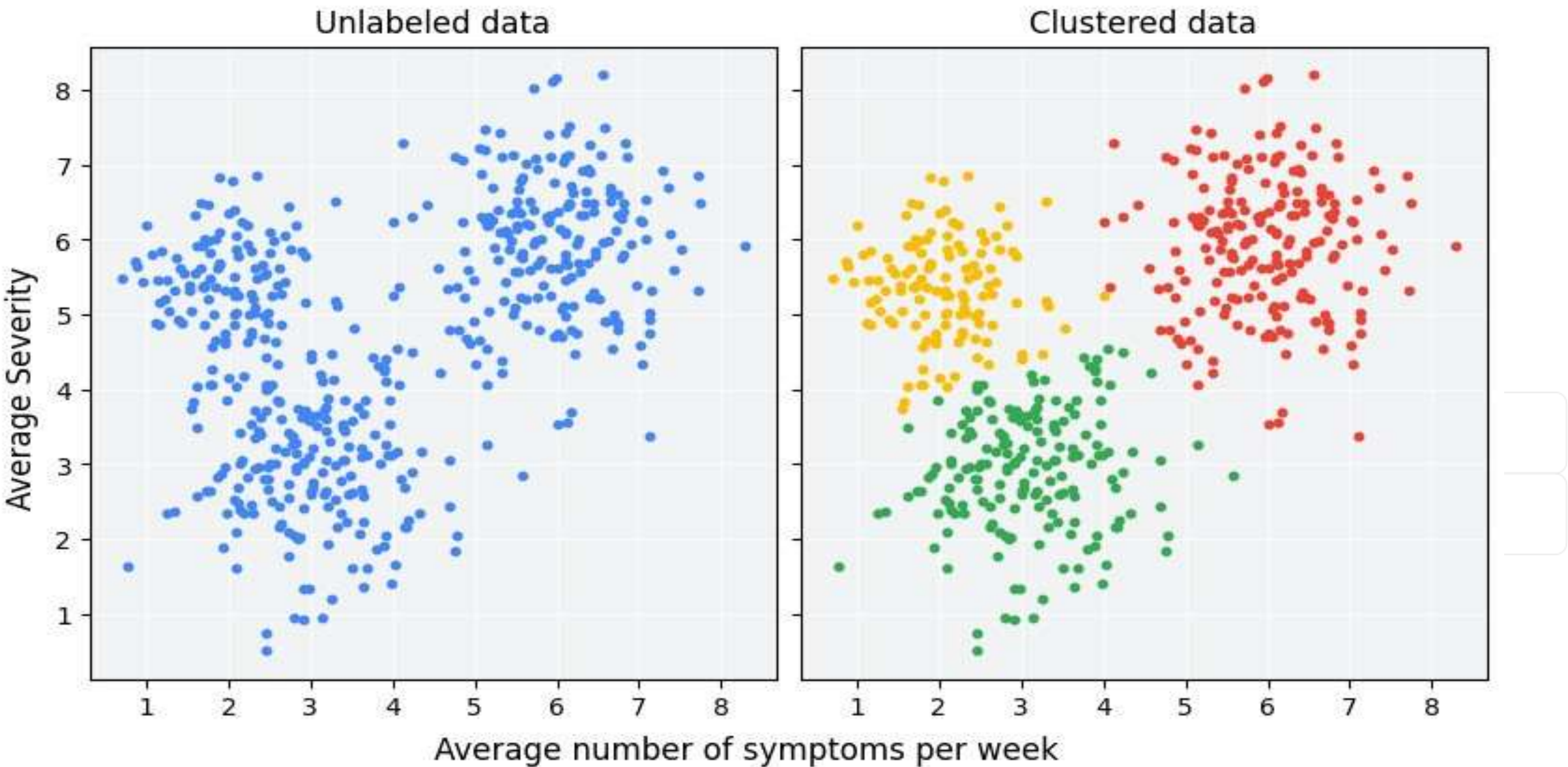
4. Association Mining

"Which items occur together?"

"Customers who buy X often buy Y."

Clustering: Finding Natural Groups

Discovering categories purely from feature characteristics.



Anomalies & Dimensionality

ANOMALY DETECTION

Identifies Extreme Outliers

Detects observations operating outside expected density bounds.

Applications: Fraud detection, system intrusions, equipment failures.

DIMENSIONALITY REDUCTION

Simplifies Complex Features

Projects 100+ feature dimensions into readable 2D or 3D spaces.

Key Method: Principal Component Analysis (PCA).

Unsupervised Toolkit

Core algorithms categorized by primary function.

CLUSTERING

K-Means: Centroid distance

DBSCAN: Density spatial regions

Hierarchical: Nested tree structures

DIMENSIONALITY

PCA: Linear variance projection *t*-

SNE: Non-linear cluster map

UMAP: Manifold structure mapping

ASSOCIATION

Apriori: Frequent itemset rules

FP-Growth: Tree-based pattern mining

Real-World Applications



Customer Segmentation



Fraud Detection



Biological Genomics



Defect Spotting



Recommendation Discovery

Unsupervised learning unlocks insights from vast pools of unannotated data across modern industries.

Reinforcement Learning (RL)

Learning through interaction, trial, and error
feedback to maximize cumulative reward.

AGENT



ACTION

ENVIRONMENT



REWARD

 *Continuous Decision Feedback Loop*

FOUNDATIONAL CONCEPT

Learning Through Trial & Reward

An agent learns to make optimal decisions by taking actions in an environment to achieve a goal.

INTUITIVE ANALOGY

Learning a Video Game

- 🕒 Try a new controller input
- 🕒 Observe score change or crash
- 🕒 Gain points (+Reward) or lose lives (-Penalty)
- 🕒 Refine strategy for next attempt

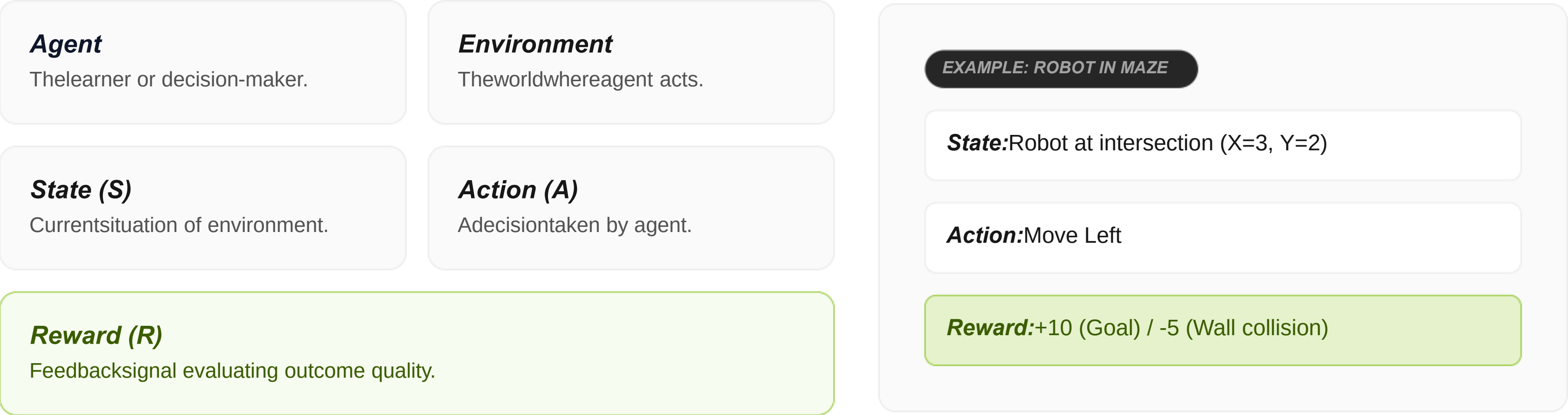
KEY DISTINCTION

No Explicit Teacher

Unlike Supervised Learning, no supervisor provides correct step-by-step labels. The agent evaluates its own choices over time.

🐱 **Learns directly from outcome consequences.**

The RL Interaction Loop



From Chaos to Optimal Strategy

1

Early Stage

Random actions & frequent mistakes.

LOW REWARDS

2

Mid Stage

Identifies rewarding patterns & reduces errors.

GROWING RETURNS

3

Late Stage

Mastered policy & highly efficient execution.

OPTIMAL REWARD

Observe State → Choose Action → Receive Reward → Update Strategy

Exploration vs. Exploitation

EXPLORATION

Try Something New

Experiment with unfamiliar actions to discover better strategies.

EXPLOITATION

Use What Works

Leverage existing knowledge to secure guaranteed high rewards.

Core Building Blocks

How the agent decides and evaluates.



Policy (π)

The Decision Strategy

Maps current environment states to the best action to take.

Question answered: "What should I do right now?"



Value Function (V/Q)

Long-Term Expected Return

Estimates total cumulative reward starting from a specific state.

Question answered: "How good is this position long-term?"

Real-World Impact & Paradigm Comparison



Autonomous Driving



Robotics Control



LLM Alignment (RLHF)



Game AI & Strategy

Paradigm	Data Type	Feedback Mechanism	Primary Goal
Supervised	Labeled examples	Direct ground truth labels	Predict target output
Unsupervised	Unlabeled raw data	No feedback signal	Discover hidden structure
Reinforcement	Environment interactions	Rewards & penalties	Maximize cumulative reward